

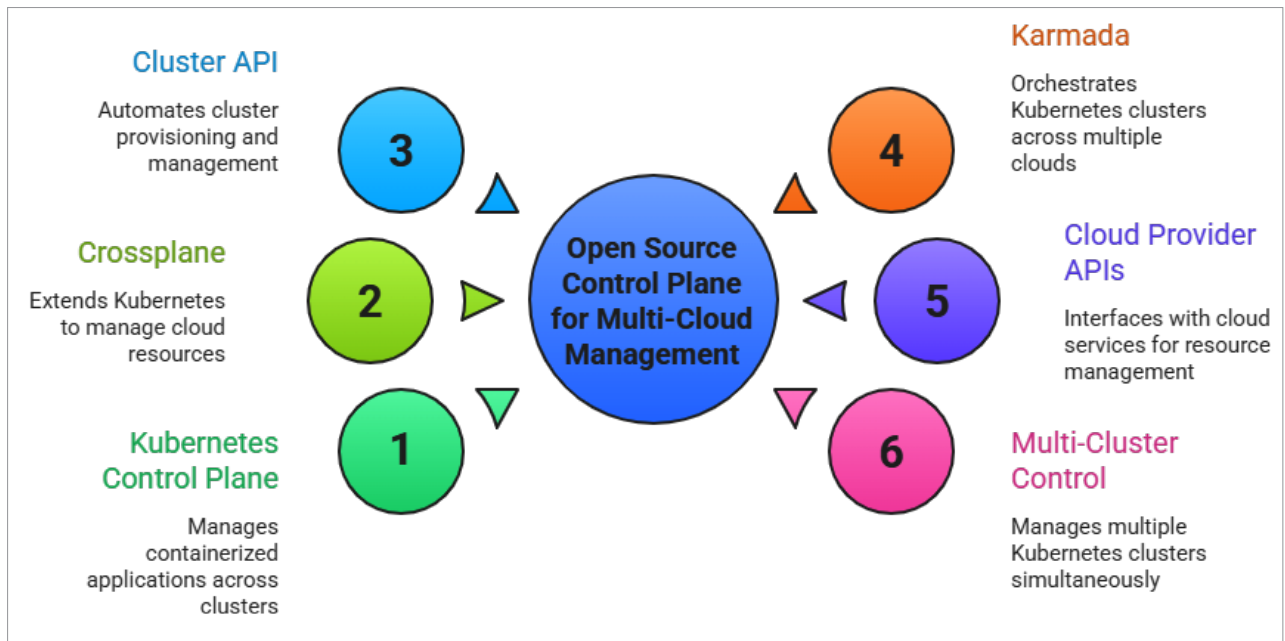


Figure 2: Open source control planes

Crossplane is widely thought of as one of the best. It enhances the features of Kubernetes by providing an expanded definition of infrastructure to include Kubernetes-resourced Infrastructure as Code.

By utilising Crossplane, organisations can provision a multitude of infrastructure elements like cloud storage buckets, databases, and virtual networks using Kubernetes manifests. In practice, a developer can define a database instance on AWS or Google Cloud using the YAML configuration within Kubernetes.

The many advantages of Crossplane include:

- One API to manage a unified infrastructure across multiple cloud providers.
- Version control of all infrastructure definitions using Infrastructure as Code.
- Standardised infrastructure templates created by platform teams and used by developers in the application development process.

Centralised platform engineering teams are creating shared foundation components (Infrastructure as

Code) for providing developers with reusable infrastructure.

GitOps for multi-cloud infrastructure

It's critical for every organisation to manage its infrastructure across different environments — automating and making its processes reproducible is essential.

The GitOps model provides tremendous value here.

GitOps is a DevOps practice that uses Git repositories to store infrastructure code for configuration, which is automatically propagated to systems via a continuous delivery pipeline. Three of the most common open source GitOps solutions are:

- Argo CD
- Flux CD
- JenkinsX

By leveraging these tools, organisations can automatically provision and upgrade resources contained within their Git repositories via CI/CD pipelines across multiple Kubernetes clusters or in different clouds.

The primary benefits of GitOps include:

- A version-controlled infrastructure

that provides historical tracking of all changes to your infrastructure.

- Docs for the deployment of infrastructure can be automated; so the infrastructure is deployed using automated CI/CD pipelines.
- Increased reliability — if there is a difference between the desired and the actual state of a given system, GitOps tools reconcile between the two states, restoring the configuration to what it was previously.

By combining GitOps with Kubernetes control planes, organisations can create an organisation-wide multi-cloud infrastructure management system that is entirely automated.

Observability and networking across multi-cloud clusters

For monitoring several workloads in different clouds, a singular observable stack is of great importance. There are many open source tools available for monitoring in Kubernetes-based environments, but the most common include:

- Prometheus
- Grafana